When Does Spectral Shrinkage Improve Clustering? A Reproduction and Failure Analysis of SSC

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Abstract

Spectral clustering embeds points with Laplacian eigenvectors and discretizes with K-means. Spectral Shrunk Clustering (SSC) of Chang, Nie, Ma, Yang, and Zhou then shrinks that embedding on a second graph before the same K-means. This COMP3314 report reimplements SSC from the printed equations, without author code, and asks when that shrinkage improves clustering. Variant E8 is the stationary system of Equation (8); printed Equation (13) matches the same objective with graph coefficient $\gamma/2$. E8 is a reconstruction, not a claim about the authors' historical binary. On AT&T ORL, Zenodo JAFFE, cropped YaleB, and LIBSVM USPS, with k=5, protocol γ=1, and n=50 random K-means trials, the published SC/SSC table is not reproduced. Paired ΔACC at γ=1 is about $+1.1$ points on ORL and negative on YaleB. Shrinkage does not stably help. The YaleB gap is already present in K-means on F. k-means++ is a Level C diagnostic. SPLS is unimplemented. The project is not marked fully reproduced.

1 Introduction

Spectral clustering builds a similarity graph, takes Laplacian eigenvectors, and runs K-means in that embedding [Shi and Malik, 2000, von Luxburg, 2007]. COMP3314 contrasts that neighbourhood geometry with PCA, which keeps high-variance axes, and with LDA, which uses class scatter. Chang et al. argue that a graph on the original points x_i need not be the right regularizer for the embedded points f_i. SSC therefore computes a normalized-cut embedding F, rebuilds a k-nearest-neighbour (kNN) graph on F, and solves a convex shrinkage problem for G before the final K-means [Chang et al., 2015].

This is a paper study with critical analysis, not a leaderboard. We reimplement SSC from the equations, tag unspecified choices as A/B/C, obtain datasets only from files with a URL and checksum, and never copy a published ACC/NMI into an Our Reproduction cell. A closer number does not promote a diagnostic to the paper method. No named dataset is marked fully protocol-reproduced.

2 SSC objective and the E8/E13 split

Lloyd K-means is the last discretization in SSC and in the spectral baseline [Lloyd, 1982]. The paper uses the symmetric normalized Laplacian of Equation (4) and sets k=5 for every spectral method. We take δ^2 as the median of positive directed kNN squared distances (unspecified; Category B) and undirected edges as the union of the two directed relations (Category B; von Luxburg, 2007). F holds

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the c eigenvectors for the smallest eigenvalues. Eigenvectors are *not* row-normalized; Ng–Jordan–Weiss row-norm is Category C even when ACC is higher [Ng et al., 2001]. Pattern shrinking is cited as SPLS; we did not obtain Section 3 of that paper and keep SPLS as `NOT_IMPLEMENTED` [Hou et al., 2013]. LDA-SC is named but not defined; it stays `APPROXIMATE` and is omitted from Table 1.

A second Gaussian kNN graph W is built on the rows of F. The printed convex problem is

$$\min_G \|G - F\|_{2,1} + \gamma \sum_{ij} W_{ij} \|g_i - g_j\|_2, \tag{1}$$

with $\|H\|_{2,1} = \sum_i \|h_i\|_2$ over rows and the sum over ordered pairs. Convexity is only in G given fixed F, W with W nonnegative. It does not make spectral clustering or K-means convex.

E8 versus E13. Differentiating (1) when both ordered pairs of a symmetric W are stored yields the stationary system $(S + 2\gamma\widetilde{L})G = SF$. That linear system is variant **E8**. Printed Equation (13) uses $\gamma\widetilde{L}$ instead of $2\gamma\widetilde{L}$: it is the stationary point of (1) with graph coefficient $\gamma/2$, not of Equation (8) at the printed γ. Algorithm 1 line 10 replaces $\widetilde{L}$ by $\widetilde{W}$ and F by X; we do not execute that line. E8 is the primary reconstruction of Equation (8). It is *not* claimed to be the authors' historical code. An existing small-graph CVXPY/CLARABEL check on an n=6 chain (γ=0.8, seed 1) is consistent with this split: E8 matches Equation (8) at the printed γ (max $|G|$ gap 1.17×10^{-5}); E13 matches the $\gamma/2$ instance (max $|G|$ gap 2.33×10^{-5}). A pass of that test is not a new proof.

Default G_0=F. We stop on a relative Frobenius step of 10^{-6} or after 300 iterations. Labels come from K-means on G and are never written back into W or F.

3 Paper evidence versus our protocol

Published Table 1 reports mean±std ACC/NMI. The experimental text also says each clustering is repeated 50 times with random initialization and the reported result is the one with the best objective. Those two sentences are not the same protocol. Our ± is the sample SD over 50 distinct K-means seeds on one embedding (E6): F and G are solved once.

Four datasets have a completed n=50 protocol. ORL is the AT&T 92×112 release resized bilinearly to 32×32 (`att92`), not an unidentified third crop [Samaria and Harter, 1994]. JAFFE is the verified Zenodo TIFF set, 213 images, resized to 26×26 [Lyons et al., 1998, 2025]. YaleB is CroppedYale 2414 non-ambient $(192, 168)$ images resized bilinearly to 32×32 (`cropped2414`), not claimed pixel-identical to Deng Cai's mat [Georghiades et al., 2001, Lee et al., 2005]. USPS is the LIBSVM 9298×256 pair [Hull, 1994]. Counts match the paper for JAFFE, YaleB $(2414/38)$, and USPS. AR was not downloaded; UMIST archives yield 565 unique cropped images, not 564 [Graham and Allinson, 1998]; MSRA50 was not found. Those three names are blocked and are not filled as `EXACT`.

Shared `SpectralGraphConfig`: k=5, union kNN, median directed bandwidth, Equation (4), no row-norm. Primary SSC cell: variant E8, γ=1 frozen without labels (not selected by ACC), G_0=F, IRLS budget 300, K-means `init="random"`, n_{init}=1, seeds 0. . .49. The γ grid $\{10^{-6}, 10^{-3}, 1, 10^3, 10^6\}$ is reported separately. ACC is Hungarian matching [Munkres, 1957]; NMI is the Strehl–Ghosh geometric mean [Strehl and Ghosh, 2002]. Code: `https://github.com/SelinaMai-SYM/COMP-3314-G23.git` branch `v1`. Records: `results/strict_reproduction_matrix.csv` and `results/final_round/`.

4 Main comparison

Table 1 is the only main-text comparison. Paper columns quote Chang et al. and are labelled historical only. Our SC and E8 cells are the n=50 strict-paper runs, not Round 2 pilots with YaleB n=10 or USPS n=3.

5 Critique

Convexity is not clustering quality. Equation (8) is convex in G given fixed F, W. That does not certify a unique clustering metric. On three real-data solves that had not met the step rule at

Table 1: Hungarian ACC and geometric NMI (%). Paper cells are reported in Chang et al. (2015), historical only. Our SC and E8 γ=1 are n=50 from `strict_reproduction_matrix.csv`. k-means++ and row-norm are not in this table.

Set (n=50)	Metric	Paper SC	Our SC	Paper SSC	Our E8 γ=1
ORL att92	ACC	72.7±2.3	56.2±4.2	76.0±1.6	57.3±3.8
	NMI	85.8±1.9	80.5±2.3	88.6±1.5	81.1±2.2
JAFFE zenodo	ACC	76.1±1.6	68.5±11.2	81.2±1.6	68.4±11.1
	NMI	80.2±0.9	83.0±5.7	84.3±1.6	83.0±5.7
YaleB cropped	ACC	34.8±1.4	18.3±1.3	49.8±1.6	17.6±1.2
	NMI	55.6±1.6	26.9±0.7	46.8±1.3	26.2±0.9
USPS libsvm	ACC	64.3±1.4	61.7±8.1	75.5±1.9	61.7±8.1
	NMI	69.5±1.6	74.9±4.5	79.8±1.6	75.2±4.6

Mean±sample SD (ddof=1) over n=50 random K-means trials (seeds 0. . .49, n_{init}=1) on one embedding. Paper columns are *Reported in Chang et al. (2015)*, historical only, not Our Reproduction. Protocol: Equation (4), k=5, union kNN, median directed bandwidth, raw eigenvectors, E8 γ=1, G_0=F. ORL is AT&T 92×112 bilinear 32×32 (`att92`). JAFFE is verified Zenodo (213). YaleB is CroppedYale 2414 bilinear 32×32 (`cropped2414`), not Deng Cai. USPS is LIBSVM 9298. k-means++, row-norm, LDA-SC (`APPROXIMATE`), and SPLS (`NOT_IMPLEMENTED`) are omitted. AR, UMIST-564, and MSRA50 are blocked, not filled as `EXACT`.

300 iterations (ORL γ=1, YaleB γ=1, USPS γ=1000), continuing until the rule fires (825/359/355 steps; 2000 identical) changes the objective by 2.5×10^{-5}, 1.3×10^{-5}, and 6×10^{-6}. ACC goes $0.57265\rightarrow0.57135$ (ORL, down), $0.17622\rightarrow0.17614$, and $0.73687\rightarrow0.73861$. After the step-size stop, the linear residual is still $0.002/0.171/0.058$. That residual is not a global-opt certificate. Extra IRLS past 300 does not account for the table gaps.

The gap is already in K-means on F. Table 1 SC sits below published SC before any shrinkage: ORL 56.2 vs 72.7, JAFFE 68.5 vs 76.1, YaleB 18.3 vs 34.8, USPS 61.7 vs 64.3. Paired ΔACC =ACC(G)−ACC(F) at E8 γ=1, same 50 seeds, 95% bootstrap CI reflecting only K-means noise on one fixed (F, G): ORL random $+0.0109$ $[+0.0030, +0.0187]$; JAFFE random -0.0009 (CI contains 0); YaleB random -0.0072 $[-0.0097, -0.0048]$; USPS random -0.0004 (CI contains 0). Only ORL's interval lies above zero, by about one percentage point. YaleB's interval lies below zero.

Shrinkage almost never helps at protocol γ=1. On the saved five-point grid, larger γ lowers TV(G) under the ordered-pair sum, but a larger TV drop is not higher ACC. YaleB ACC falls monotonically as γ grows (0.1834, 0.1762, 0.1705, 0.1289). USPS is flat at γ=1 (0.6170), higher at γ=1000 (0.7369), and collapses at γ=10^6 (0.3881). That γ=1000 cell was not chosen by maximizing ACC and is not "the" reproduced USPS number. The paper's text already treats USPS as an exception to γ=1. We do not select γ by ACC, and we do not conclude that the paper is wrong because our ACC is lower.

YaleB data versions. Cropped2414 and Deng Cai are both 2414×1024, 38 classes, pixels in $[0, 1]$, with the same per-class counts. Best alignment (transpose) has median L2 1.92; zero images coincide at 10^{-8}. Under the identical graph rule and seeds, K-means(F) ACC is 0.1834 (cropped) vs 0.2016 (Deng Cai). Published SC is 34.8%. The version gap does not explain that distance. Bandwidth, symmetrization, row-norm, and interpolation are B or C; the highest ACC among them is not adopted as the paper method.

k-means++ is a Level C diagnostic. Under k-means++, paired ΔACC at γ=1 is $+0.0006$ on ORL (CI contains 0) and remains negative on YaleB (-0.0023, CI below 0). The initializer lifts ORL K-means(F) from 0.562 to 0.783 and JAFFE from 0.685 to 0.953. JAFFE ++ exceeds published SSC 81.2%, so "they must have used ++" does not fit every dataset. YaleB ++ reaches only 0.218, far from 34.8%. These arms are not the paper method and are not a reproduction of the table gap.

6 Existing mutual-kNN extension

The one planned extension replaces union kNN W on F by mutual (AND) kNN with the same Gaussian weights frozen before IRLS [Brito et al., 1997, Maier et al., 2009]. Convexity of (1) is unchanged; no new method and no new theorem are claimed. On JAFFE at γ=1, k=5, n=50, mutual E8 ACC is 68.4±11.2 versus union E8 68.4±11.1 (410 vs 655 edges; 20 vs 9 components). The outcome is negative/neutral, not a win, and it does not close the table gaps.

7 Conclusion

CONFIRMED. On the $n{=}6$ chain already in the test suite, E8 matches Equation (8) at the printed γ and E13 matches $\gamma/2$. ACC is Hungarian; NMI is geometric. The $n{=}50$ cells use seeds $0\ldots49$ on one F (and one G).

SUPPORTED. At E8 $\gamma{=}1$, shrinkage is not a reliable ACC improvement over SC on these four reconstructions. ORL random init is a small positive exception; YaleB is negative. Greater shrinkage is not monotone in ACC. Finishing the IRLS step rule after iteration 300 does not raise ACC in any substantial way. Deng Cai versus cropped2414 moves YaleB SC by about 1.8 points, not from 18% to 35%. The large YaleB gap is already in K-means on F.

UNRESOLVED. The authors' linear system (line 10, Equation 13, or E8); δ, union versus mutual, and the $32{\times}32$ kernel; which YaleB pixels and which ORL files were resized; the γ behind each Table 1 SSC entry; whether printed $\pm$ is 50 K-means trials on one G or 50 full SSC solves. AR, UMIST-564, BinAlpha as a paper identity, MSRA50, and SPLS are not reproduced. k-means++ does not explain the published gap as a reproduction. This project is not marked fully reproduced.

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HKU COMP3314 course report. No external funding is claimed. Names and emails require confirmation. Code is intended for `https://github.com/SelinaMai-SYM/COMP-3314-G23.git` on branch `v1`.

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